

P YATHIN MACHAIAH

Software & Security Engineer · Distributed Systems · Python Developer
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github.com/PYatiM | leetcode.com/PYatiM | hackerrank.com/PYatiM

EDUCATION

Dayananda Sagar University

B.Tech — Computer Science & Engineering (Cyber Security)

Expected Sep 2027

GPA: 8.38 / 10.0

DSA & Problem Solving: 160+ LeetCode problems (Python) | HackerRank Python 5★ Badge

PROJECTS

APIGATE — Secure API Gateway

Python · FastAPI · OAuth2/JWT · Redis · OpenTelemetry · Prometheus · Docker Compose

- Architected a horizontally scalable, fault-tolerant API gateway with layered security middleware: OAuth2 JWT authentication, OpenAPI request validation, Redis-backed distributed rate limiting, structured audit logging, and TLS — built for high-availability production deployment.
- Applied OOP design principles (single responsibility, separation of concerns) to produce independently testable, swappable middleware modules — auth, rate limiting, validation, and observability each isolated as discrete components.
- Containerized the full distributed stack with Docker Compose; integrated OpenTelemetry distributed tracing and Prometheus metrics for real-time observability

VTP/1 — Asynchronous Authenticated Communication Protocol

Python · asyncio · Sockets · HMAC-SHA256 · Applied Cryptography

- Designed and implemented a custom asynchronous client-server protocol using HMAC-SHA256 message authentication to secure data exchange between distributed peers.
- Conducted an independent security review of the protocol's trust model and identified a critical flaw — a shared secret hardcoded into and imported directly by the client — that broke the intended client-server trust boundary.
- Built and ran a dedicated tamper-and-replay test suite to validate message integrity and freshness guarantees, informing a redesign of the key-distribution and authentication flow.

Secure Containerized Code Execution Sandbox

Python · FastAPI · Docker · Linux / Syscall Filtering

- Designed a fault-isolated execution engine routing untrusted code through ephemeral Docker containers with enforced CPU/memory limits, network isolation, and a 6-stage sanitization pipeline (Input → API → Runner → Limiter → Sanitizer → Response), mitigating RCE, fork-bomb, and privilege-escalation attacks at the OS level.

TECHNICAL SKILLS

- Languages: Python · Java · C · SQL · Bash
- Backend & Systems: FastAPI · REST APIs · OAuth2/JWT · Redis · Docker · Git · Microservices
- Security: Wireshark · Nmap · Burp Suite · OWASP Top 10 · Linux (Kali, Ubuntu)
- Cloud & Observability: AWS · OpenTelemetry · Prometheus

CERTIFICATIONS

Advanced Computer Networks — NPTEL / IIT	2025
Ethical Hacker — Cisco Networking Academy	2026
Google Cybersecurity Certificate	2026
CompTIA Security+	In Progress — Target Q3 2026

CTF COMPETITIONS & HACKATHONS

CTF Competitions

- Secured 13th Place in H7CTF, solving challenges across web exploitation, reverse engineering, and network security.
- Achieved 24th Place in Web Arena, demonstrating web application security assessment and vulnerability exploitation skills.
- Competed in Crackin 2.0, focusing on practical offensive security and problem-solving challenges.

Hackathons

- Qualified through the preliminary round of Hackathon X, advancing beyond the initial evaluation stage.
- Participated in the NASA Space Apps Challenge, collaborating on tech-driven solutions for real-world problems.